

Lecture 08

The Arm Cortex M4 Processor Architecture

1. Define Arm architecture and explain why it is widely used in embedded and mobile systems.
2. Describe the different Arm processor families (Cortex-A, Cortex-R, Cortex-M, SecurCore) and their applications.
3. Describe the features and typical applications of the Cortex-M processor series.
4. Differentiate between an Arm architecture and an Arm processor.
5. Explain the steps involved in designing an Arm-based System-on-Chip (SoC).
6. Describe the key features and performance characteristics of the Cortex-M4 processor.
7. Explain the architectural features of the Cortex-M4 processor including instruction set, pipeline, and interrupt support.
8. Describe the major components shown in the Cortex-M4 block diagram and their functions.
9. Explain the three-stage pipeline of the Cortex-M4 processor and how it improves performance.
10. Describe the register bank of the Cortex-M4 processor.
11. Explain the role and accessibility of general-purpose registers (R0–R12) in Cortex-M4.
12. Discuss the function of the Stack Pointer (R13) and explain the difference between MSP and PSP.
13. Describe the function of the Program Counter (PC) in Cortex-M4.
14. Explain the purpose of the Link Register (LR) in function calls.
15. Describe the components of the xPSR register and their functions.
16. Compare ARM and x86 architectures based on instruction set, performance, and application areas.